

Curriculum Learning for Multimedia in the Era of LLM

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Abstract

This tutorial focuses on curriculum learning (CL), an important topic in machine learning, which gains an increasing amount of attention in the research community. CL is a learning paradigm that enables machines to learn from easy data to hard data, imitating the meaningful procedure of human learning with curricula. As an easy-to-use plug-in, CL has demonstrated its power in improving the generalization capacity and convergence rate of various models in a wide range of scenarios such as computer vision, natural language processing, reinforcement learning, etc. In particular, CL can also play an important role in multimedia applications. Therefore, it is essential to introduce CL to more scholars and researchers in the machine learning and multimedia community. However, there have been no tutorials on CL for multimedia so far, motivating the organization of this tutorial at ACM Multimedia 2024. To give a comprehensive tutorial on CL for multimedia, we plan to organize it from the following aspects: (1) theories, (2) approaches, (3) applications, (4) tools, and (5) future directions. First, we introduce the motivations, theories, and insights behind CL. Second, we advocate novel, high-quality approaches, as well as innovative solutions to the challenging problems in CL. Then we present the applications of CL in various scenarios, especially multimedia, followed by some relevant tools. In the end, we discuss open questions and future directions in the era of large language models. We believe this topic is at the core of the scope of ACM Multimedia and is attractive to the audience interested in machine learning and multimedia from both academia and industry.

CCS Concepts

• **Computing methodologies** → **Machine learning; Artificial intelligence.**

Keywords

Curriculum Learning, Machine Learning Paradigm, Training Strategy, Machine Learning Library and Tool, Large Language Models

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1 Tutorial Summary

1.1 Introduction

Curriculum learning (CL) has continuously gained attention since its first advent in 2009 [1]. CL borrows the idea of human learning curricula from easy content to hard content, forming a general training strategy for various machine learning models and applications [4]. Given that CL can help models to generalize better and converge faster, researchers have proposed numerous CL algorithms and shown their effectiveness in a wide range of tasks. Therefore, we believe that it is necessary to introduce CL to more researchers in machine learning community and provide an overall picture of CL, which includes comprehensible and elaborate answers to the following questions: (1) Theories: what are the definitions of CL and why are they effective? (2) Approaches: what methods should be included? (3) Applications: how to design different curricula for different scenarios, especially multimedia? (4) Tools: what tools are available for ease of use and understanding of CL? (5) Future directions: what role can CL play in the context of multimedia and in the era of large models?

1.2 Theories

A curriculum is a sequence of training criteria over T steps: $C = \langle Q_1, \dots, Q_t, \dots, Q_T \rangle$. Each criterion Q_t is a reweighting of the target distribution $P(z)$: $Q_t(z) \propto W_t(z)P(z)$, $\forall z \in D$, where z is an example and D is the training set and the following 3 conditions are satisfied:

- The entropy of distributions increases, i.e., $H(Q_t) < H(Q_{t+1})$.
- The weights increase, i.e., $W_t(z) \leq W_{t+1}(z)$, $\forall z \in D$.
- $Q_T(z) = P(z)$.

Since the concept of *Original Curriculum Learning* was formally proposed as above, the academic community follows and further extends the definition of CL within the spirit of “training from easier data (tasks) to harder data (tasks)”, i.e., relaxing the conditions in its definition to enable more flexible CL strategies. For example, *Data-level Generalized Curriculum Learning* is defined as a sequence of reweighting of target training distribution over T training steps,

discarding the 3 conditions. *Generalized Curriculum Learning* is defined as a sequence of training criteria over T training steps, discarding the 3 conditions and the definition of Q_t .

Another important question is why this human-curriculum-like training strategy work. Basically, existing analyses uncover the essence of CL from the perspectives of *optimization problem* and *data distribution*, based on which we can summarize the two main motivations for applying CL: *to guide* and *to denoise*.

1.3 Approaches

A general framework for curriculum design consists of two core components: Difficulty Measurer and Training Scheduler, which decide two things respectively: (1) What kind of training data is supposed to be easier than other data? (2) When should we present harder data for training, and how much more?

According to these two components, we can divide existing CL methods into two types: when both the Difficulty Measurer and Training Scheduler are designed by human prior knowledge with no data-driven algorithms involved, we call the CL method predefined CL. If any (or both) of the two components are learned by data-driven models or algorithms, then we denote the CL method as automatic CL.

1.4 Applications

CL has a wide range of applications. In this tutorial, we discuss them in terms of the following aspects:

- Combinatorial optimization: traveling salesman problem [7], etc.
- Computer vision: visual concept recognition [6], etc.
- Natural language processing: machine translation, etc.
- Multimodal learning: visual grounding [2], etc.
- Recommendation: social recommendation [5], etc.
- Graph machine learning: node classification [3], etc.
- Robotics: navigation, control, games, etc.
- Other cutting-edge areas: neural architecture search, etc. [10]

1.5 Tools and Benchmarks

This tutorial will introduce our contributed Curriculum Machine Learning library, CurML [8], which is the first public open-source library for CL. We implement a considerable number of existing CL algorithms through a unified and extensive framework.

Besides, we will introduce our developed Curriculum Learning Benchmark, CurBench [9], which is the first benchmark for CL. It includes 5 datasets spanning 3 research domains, 9 backbone models, 3 training settings, and 2 evaluation dimensions, along with a toolkit for reproducing 15 core curriculum learning methods

1.6 Future Directions

Specialized datasets. In existing literature, CL has been applied to diverse tasks and domains. It is necessary but challenging to design a specialized dataset with unified metrics to evaluate and compare the CL algorithms.

More advanced theories. Existing theoretical analyses provide different angles for understanding CL. Nevertheless, more theories are still required to guarantee the effectiveness of CL, and the application of CL in a specific task.

Application on large models. Previous research on CL has mainly focused on smaller models. But with the rise of large models in today's landscape, there's a pressing question: How can we effectively combine CL with large models? Specifically, can CL help with tasks like pre-training, fine-tuning, and prompting in large language and multimodal models, ultimately speeding up learning, improving generalization, and making the models more adaptable to new tasks or domains? We'll provide some examples and analysis.

1.7 Q&A

This tutorial includes 15 minutes for questioning and answering. We welcome any question about CL and multimedia from the audiences.

2 Target Audience

This tutorial will be highly accessible to the whole machine learning and multimedia community, including researchers, scholars, engineers, and students with related backgrounds on computer vision, natural language processing, graph machine learning, multimodal learning, etc. This tutorial is self-contained and designed for introductory and intermediate audiences. No special prerequisite knowledge is required to attend this tutorial.

3 Tutorial Material

This tutorial provides detailed materials from a particularly relevant IEEE TPAMI survey by the organizers (<https://www.computer.org/csdl/journal/tp/2022/09/09392296/1sq7rtKfyyQ>) and an open-source code library and benchmark by the organizers for hands-on coding demonstrations (<https://github.com/THUMNLab/CurML> and <https://github.com/THUMNLab/CurBench>).

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